VI SEMESTER (OPEN ELECTIVE-II)	L	T	P	C
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20EE6O01 - FUNDAMENTALS OF ELECTRICAL VEHICLE				

COURSE OUTCOMES:

After successful completion of this course, students should be able to:

CO1: Illustrate different types of electric vehicles

CO2: Select suitable power converters for EV applications

CO3: Design HEV configuration for a specific application

CO4: Choose an effective method for EV and HEV applications

CO5: Analyse a battery management system for EV and HEV

UNIT-I: INTRODUCTION

Fundamentals of vehicles - Components of conventional vehicles - drawbacks of conventional vehicles – Need for electric vehicles - History of Electric Vehicles – Types of Electric Vehicles – Advantages and applications of Electric Vehicles.

UNIT-II: COMPONENTS OF ELECTRIC VEHICLES

Main components of Electric Vehicles – Power Converters - Controller and Electric Traction Motor – Rectifiers used in EVs – Bidirectional DC–DC Converters – Voltage Source Inverters – PWM inverters used in EVs.

UNIT-III: HYBRID ELECTRIC VEHICLES

Evolution of Hybrid Electric Vehicles – Advantages and Applications of Hybrid Electric Vehicles – Architecture of HEVs - Series and Parallel HEVs – Complex HEVs – Range extended HEVs – Examples - Merits and Demerits.

UNIT-IV: MOTORS FOR ELECTRIC VEHICLES

Characteristics of traction drive - requirements of electric machines for EVs – Different motors suitable for Electric and Hybrid Vehicles – Induction Motors – Synchronous Motors – Permanent Magnetic Synchronous Motors – Brushless DC Motors – Switched Reluctance Motors (Construction details and working only).

UNIT-V: ENERGY SOURCES FOR ELECTRIC VEHICLES

Batteries - Types of Batteries – Lithium-ion - Nickel-metal hydride - Lead-acid – Comparison of Batteries - Battery Management System – Ultra capacitors – Flywheels – Fuel Cell – it's working.

Text Books:

1. Iqbal Hussein - Electric and Hybrid Vehicles: Design Fundamentals - CRC Press - 2021.
2. Denton - Tom. Electric and hybrid vehicles. Routledge - 2020.

Reference Books:

1. Kumar - L. Ashok - and S. Albert Alexander. Power Converters for Electric Vehicles. CRC Press - 2020.
2. Chau - Kwok Tong. Electric vehicle machines and drives: design - analysis and application. John Wiley & Sons - 2015.
3. Berg - Helena. Batteries for electric vehicles: materials and electrochemistry. Cambridge university press - 2015.